

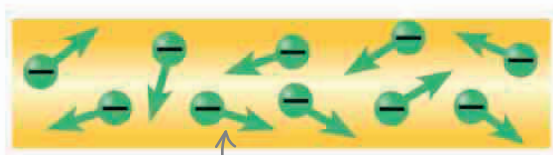
Current, voltage, and resistance

THESE ARE THE FACTORS THAT DETERMINE HOW ELECTRICITY FLOWS THROUGH A CIRCUIT.

There are two variables that control the amount of current that flows around a circuit: voltage and electrical resistance.

What is voltage?

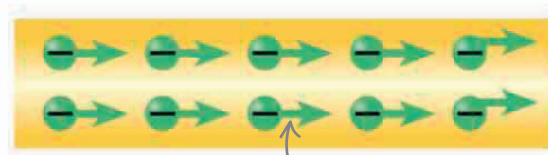
Voltage is a measure in volts (V) of potential difference—the difference in electrical energy between two points, such as the difference in potential energy at two different points of a circuit. A voltage is required to make electrons move and an electric current to flow. Batteries are labeled in terms of their voltage. A typical car battery is 12 volts, while a flashlight battery may be 1.5 volts.



electrons move in all directions

△ No voltage

If a conductor's ends are not connected to a battery or other power source, the free electrons within it drift randomly in all directions.



electrons drift in the same direction

△ Voltage

If the ends of a conductor are connected to a battery, the battery's voltage makes electrons drift along, creating electrical current.

SEE ALSO

◀ 112 What is an ion?
 ◀ 148–149 Electrochemistry
 ◀ 168–169 Inside atoms
 ◀ 203 Electric currents

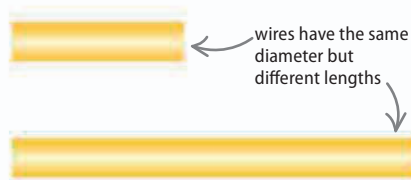
Circuits

206–207 ▶

Volts, amperes, and ohms are named after three scientists who helped develop the **science of electricity**: Alessandro Volta, André-Marie Ampère, and Georg Simon Ohm.

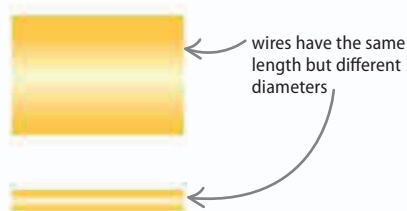
Resistance

Any piece of wire and any component in a circuit holds back the flow of electricity through it to some extent. This is usually because electrons moving around the circuit are scattered ("bounced") by the ions (charged atoms) of the material, which slows the electrons down and makes them lose energy. This "holding back" is called electrical resistance. The lost energy appears in the form of heat, sound, or light. The resistance of a wire depends on factors such as its length and diameter.



◁ Length

A shorter wire has less resistance than a longer wire of the same diameter. This is because electrons have less distance to travel and suffer fewer collisions and energy loss. In a longer wire, they have farther to go, so they encounter more collisions, greater resistance, and greater energy loss.



◁ Diameter

A thinner wire has a greater resistance than a thicker wire of the same length, because it has less room for electrons to move through. In the thicker wire, more electrons can travel side by side (like a crowd in a wide corridor), so the electron flow is greater.